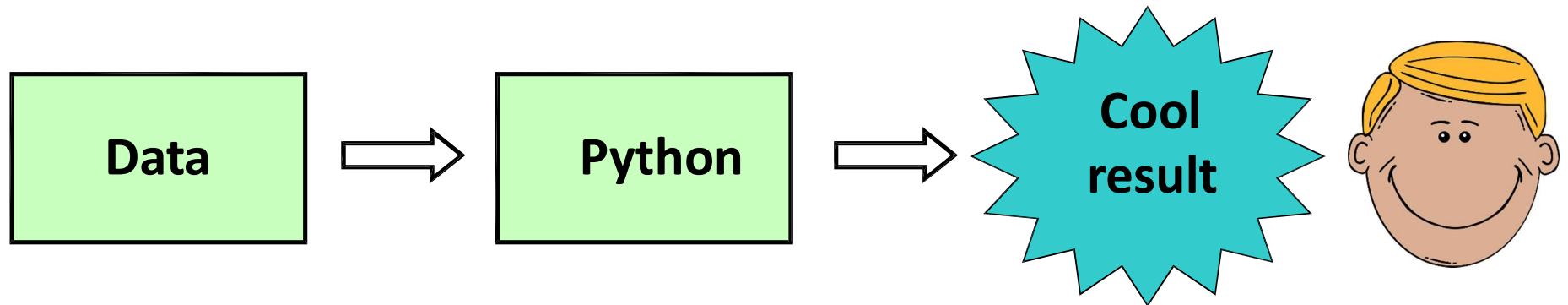
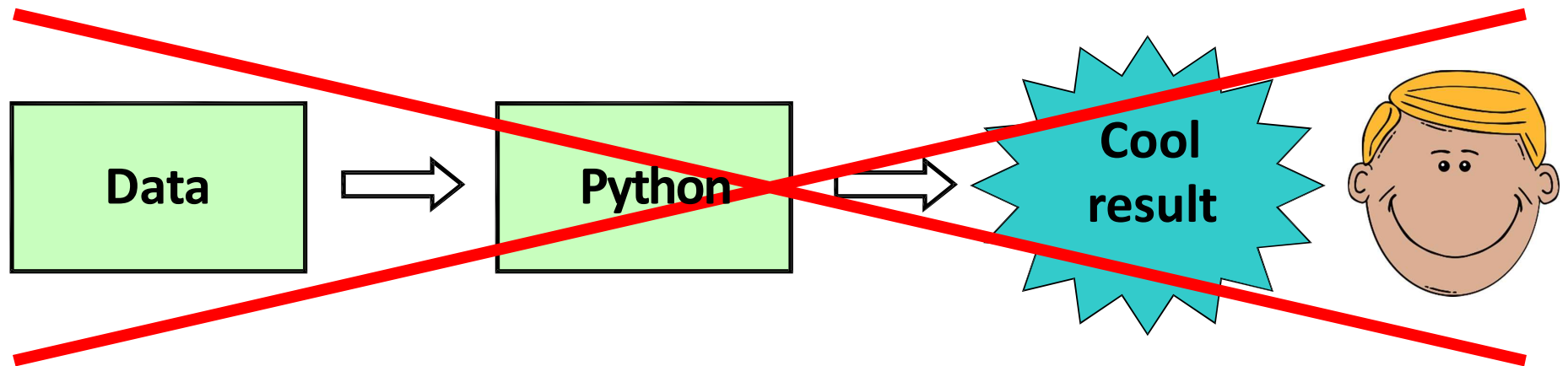

The data mining process

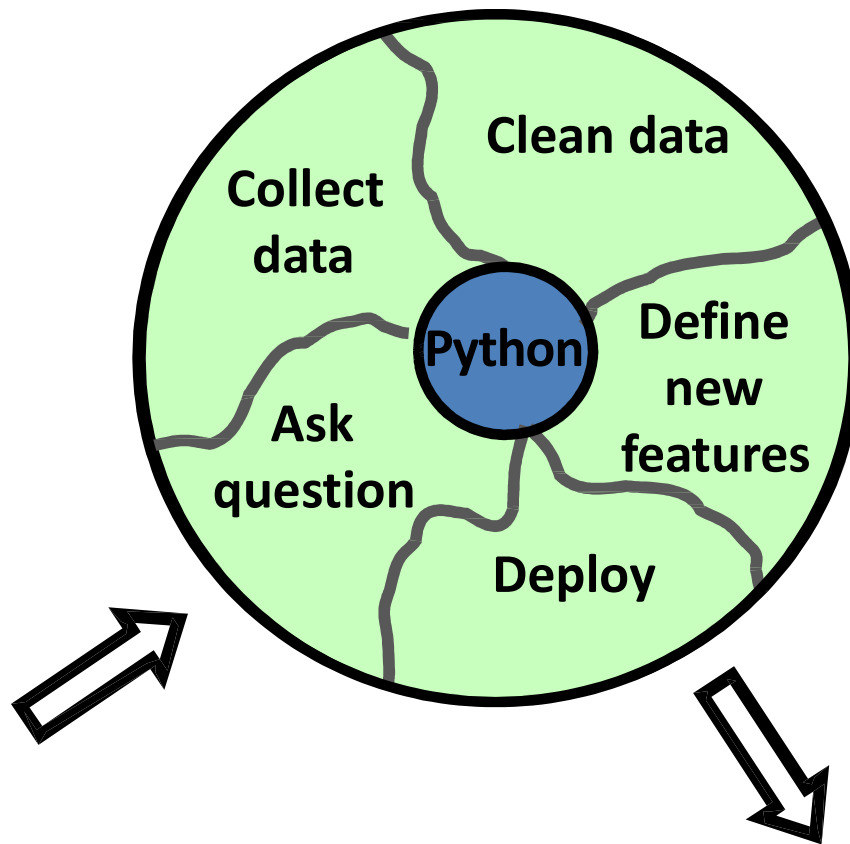
The data mining process



The data mining process



The data mining process



The data mining process

- ❖ Ask a question
 - *what do you want to know?*
 - *“tell me something cool about the data” is not enough!*
- ❖ Gather data
 - *there’s soooo much around ...*
 - *... **but** ... we need (expert?) classifications more data beats*
 - *a clever algorithm*
- ❖ Clean the data
 - *real data is very mucky*
- ❖ Define new features
 - *feature engineering—the key to data mining*
- ❖ Deploy the result
 - *technical implementation*
 - *convince your boss!*

The data mining process

(Selected) filters for feature engineering

- ❖ AddExpression (MathExpression)
 - Apply a math expression to existing attributes to create new one (or modify existing one)*
- ❖ Center (Normalize) (Standardize)
 - Transform numeric attributes to have zero mean (or into a given numeric range) (or to have zero mean and unit variance)*
- ❖ Discretize (also supervised discretization)
 - Discretize numeric attributes to have nominal values*
- ❖ PrincipalComponents
 - Perform a principal components analysis/transformation of the data*
- ❖ RemoveUseless
 - Remove attributes that do not vary at all, or vary too much*
- ❖ TimeSeriesDelta, TimeSeriesTranslate
 - Replace attribute values with successive differences between this instance and the next*

Pitfalls and pratfalls

Pitfall: A hidden or unsuspected danger or difficulty

Pratfall: A stupid and humiliating action

Pitfalls and pratfalls

Be skeptical

- ❖ In data mining, it's very easy to cheat
 - *whether consciously or unconsciously*
- ❖ For reliable tests, use a completely fresh sample of data that has never been seen before

Overfitting has many faces

- ❖ Don't test on the training set (of course!)
- ❖ Data that has been used for development (in any way) is tainted
- ❖ Leave some evaluation data aside for the *very* end

Pitfalls and pratfalls

Missing values

“Missing” means what ...

- ❖ Unknown?
- ❖ Unrecorded?
- ❖ Irrelevant?

Should you: 1. Omit instances where the attribute value is missing?
or 2. Treat “missing” as a separate possible value?

Is there significance in the fact that a value is missing?

Most learning algorithms deal with missing values

– but they may make different assumptions about them

Pitfalls and pratfalls

No free lunch



- ❖ 2-class problem with 100 binary attributes
- ❖ Say you know a million instances, and their classes (training set)
- ❖ You don't know the classes of $2^{100} - 10^6$ examples!
(that's 99.9999...% of the data set)
- ❖ How could you possibly figure them out?

In order to generalize, every learner must embody some knowledge or assumptions beyond the data it's given

A learning algorithm implicitly provides a set of assumptions
There can be no “universal” best algorithm (no free lunch)

Data mining is an experimental science

Pitfalls and pratfalls

- ❖ Be skeptical
- ❖ Overfitting has many faces
- ❖ Missing values – different assumptions
- ❖ No “universal” best learning algorithm
- ❖ Data mining is an experimental science
- ❖ It's very easy to be misled

Information privacy laws (in Europe, but not US)

- ❖ A purpose must be stated for any personal information collected
- ❖ Such information must not be disclosed to others without consent
- ❖ Records kept on individuals must be accurate and up to date
- ❖ To ensure accuracy, individuals should be able to review data about themselves
- ❖ Data must be deleted when it is no longer needed for the stated purpose
- ❖ Personal information must not be transmitted to locations where equivalent data protection cannot be assured
- ❖ Some data is too sensitive to be collected, except in extreme circumstances (e.g., sexual orientation, religion)

Anonymization is harder than you think

When Massachusetts released medical records summarizing every state employee's hospital record in the mid-1990s, the governor gave a public assurance that it had been anonymized by removing all identifying information such as name, address, and social security number. He was surprised to receive his own health records (which included diagnoses and prescriptions) in the mail.

Data mining and ethics

Reidentification techniques. Using publicly available records:

- ❖ 50% of Americans can be identified from city, birth date, and sex
- ❖ 85% can be identified if you include the 5-digit zip code as well

Netflix movie database: 100 million records of movie ratings (1–5)

- ❖ Can identify 99% of people in the database if you know their ratings for 6 movies and approximately when they saw the movies ($\pm$ one week)
- ❖ Can identify 70% if you know their ratings for 2 movies and roughly when they saw them

Data mining and ethics

The purpose of data mining is to discriminate ...

- ❖ who gets the loan
- ❖ who gets the special offer

Certain kinds of discrimination are unethical, and illegal

- ❖ racial, sexual, religious, ...

The purpose of data mining is to discriminate ...

- ❖ who gets the loan
- ❖ who gets the special offer

Certain kinds of discrimination are unethical, and illegal

- ❖ racial, sexual, religious, ...

But it depends on the context

- ❖ sexual discrimination is usually illegal
- ❖ ... except for doctors, who are *expected* to take gender into account

... and information that appears innocuous may not be

- ❖ ZIP code correlates with race
- ❖ membership of certain organizations correlates with gender

Correlation does not imply causation

As icecream sales increase, so does the rate of drownings.

Therefore icecream consumption causes drowning???

Data mining reveals correlation, not causation

but really, we want to predict the effects of our actions

Summary

- ❖ **There's no magic in data mining**
 - *Instead, a huge array of alternative techniques*
- ❖ **There's no single universal “best method”**
 - *It's an experimental science!*
 - *What works best on your problem?*
- ❖ **Python makes it easy**
 - *... maybe too easy?*
- ❖ **There are many pitfalls**
 - *You need to understand what you're doing!*
- ❖ **Focus on evaluation ... and significance**
 - *Different algorithms differ in performance – but is it significant?*

Data Analysis

What is Data?

Collection of data objects and their attributes

An attribute is a property or characteristic of an object

Examples: eye color of a person, temperature, etc.

Attribute is also known as variable, field, characteristic, or feature

A collection of attributes describe an object

Object is also known as record, point, case, sample, entity, or instance

Attributes

Objects

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Attribute Values

Attribute values are numbers or symbols assigned to an attribute

Distinction between attributes and attribute values

Same attribute can be mapped to different attribute values

Example: height can be measured in feet or meters

Different attributes can be mapped to the same set of values

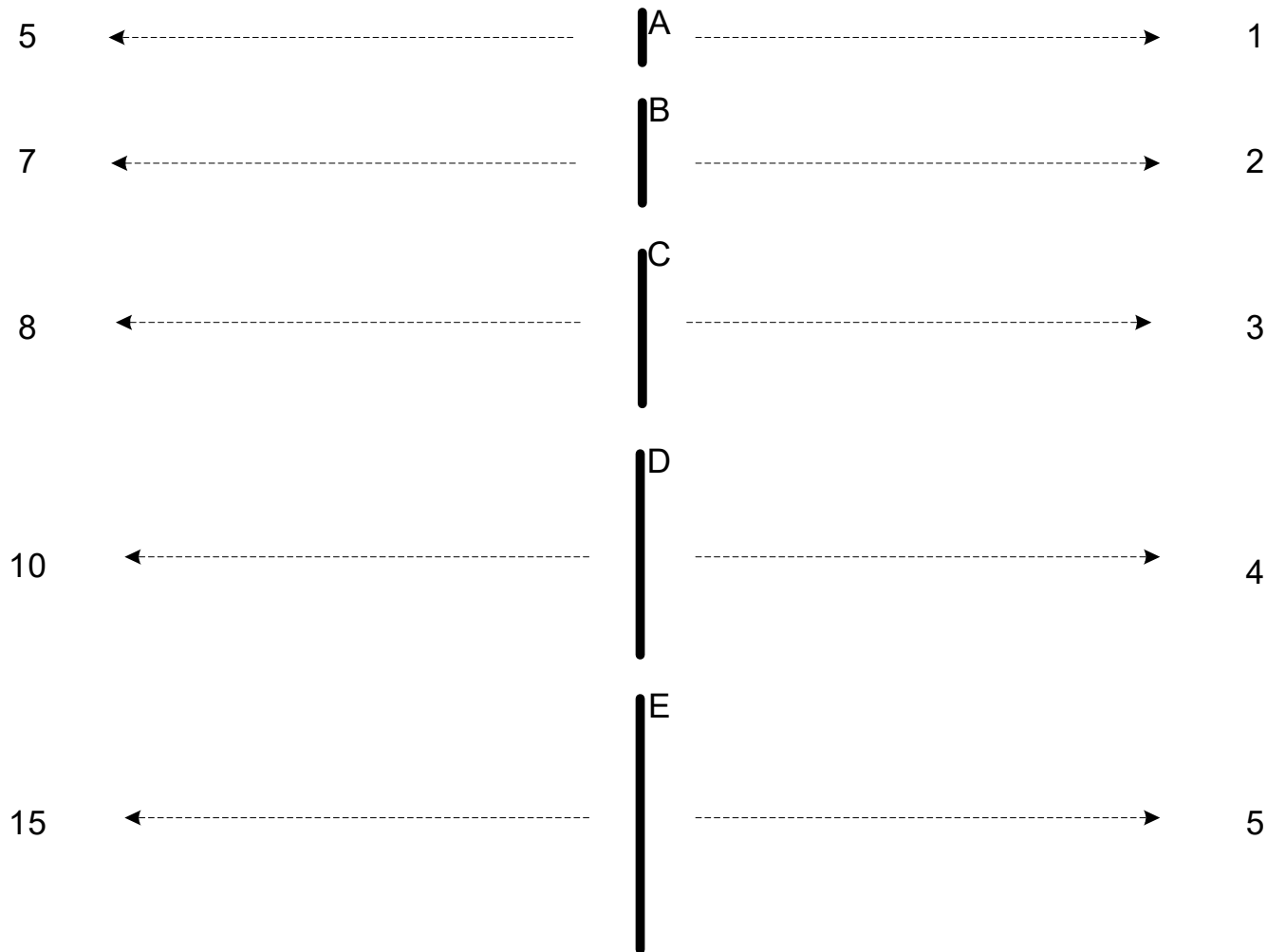
Example: Attribute values for ID and age are integers

But properties of attribute values can be different

ID has no limit but age has a maximum and minimum value

Measurement of Length

The way you measure an attribute is somewhat may not match the attributes properties.



Types of Attributes

There are different types of attributes

Nominal

Examples: ID numbers, eye color, zip codes

Ordinal

Examples: rankings (e.g., taste of potato chips on a scale from 1-10), grades, height in {tall, medium, short}

Interval

Examples: calendar dates, temperatures in Celsius or Fahrenheit.

Ratio

Examples: temperature in Kelvin, length, time, counts

Properties of Attribute Values

The type of an attribute depends on which of the following properties it possesses:

Distinctness: $= \neq$

Order: $< >$

Addition: $+ -$

Multiplication: $* /$

Nominal attribute: distinctness

Ordinal attribute: distinctness & order

Interval attribute: distinctness, order & addition

Ratio attribute: all 4 properties

Data Quality

What kinds of data quality problems?

How can we detect problems with the data?

What can we do about these problems?

Examples of data quality problems:

- Noise and outliers

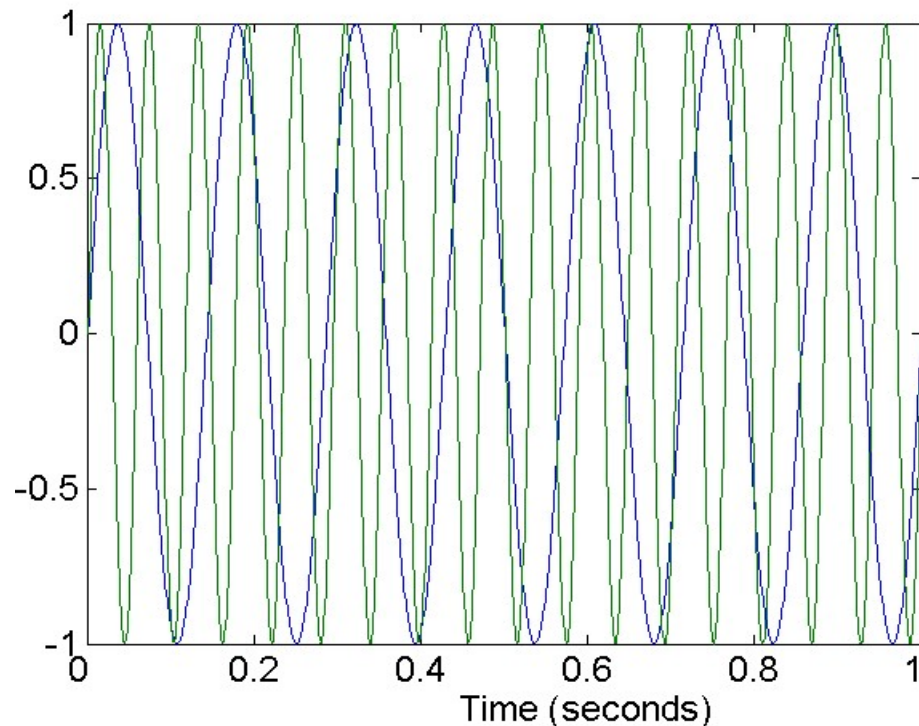
- missing values

- duplicate data

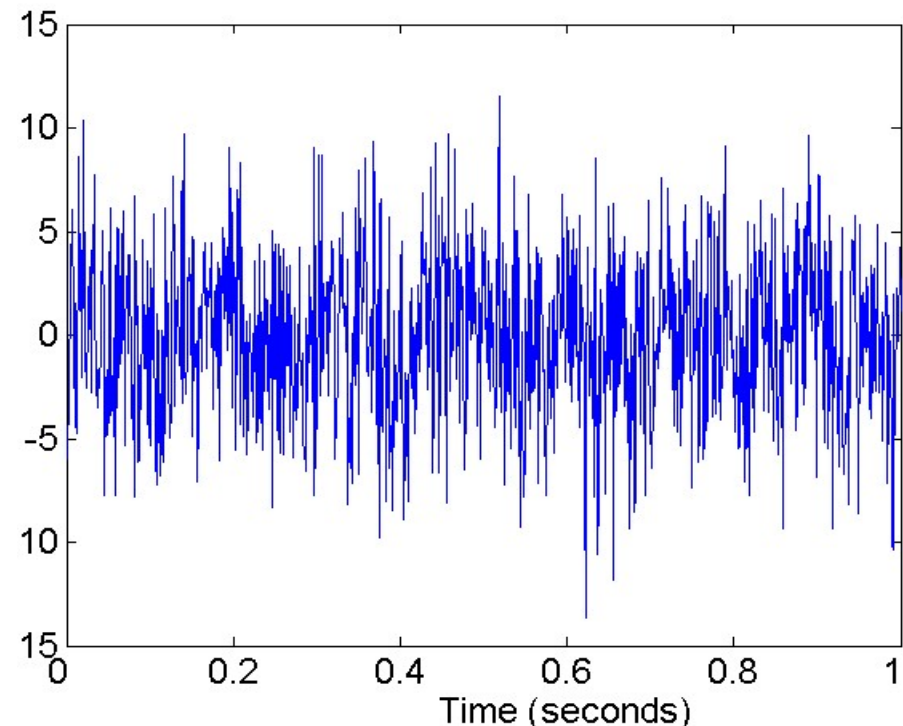
Noise

Noise refers to modification of original values

Examples: distortion of a person's voice when talking on a poor phone and “snow” on television screen



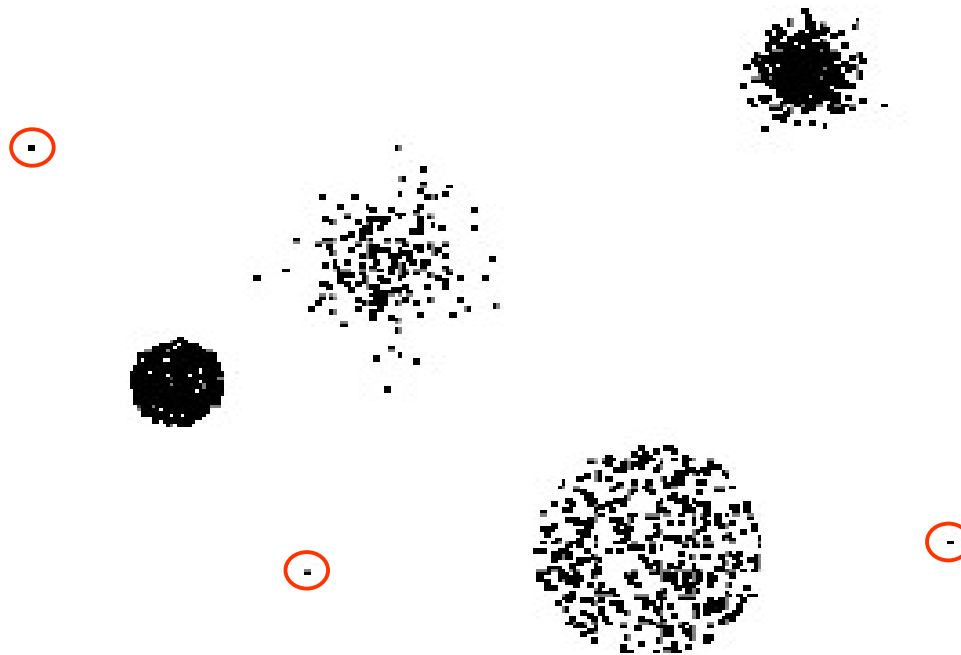
Two Sine Waves



Two Sine Waves + Noise

Outliers

Outliers are data objects with characteristics that are considerably different than most of the other data objects in the data set



Missing Values

Reasons for missing values

- Information is not collected
(e.g., people decline to give their age and weight)
- Attributes may not be applicable to all cases
(e.g., annual income is not applicable to children)

Handling missing values

- Eliminate Data Objects
- Estimate Missing Values
- Ignore the Missing Value During Analysis
- Replace with all possible values (weighted by their probabilities)

Duplicate Data

Data set may include data objects that are duplicates, or almost duplicates of one another

Major issue when merging data from heterogeneous sources

Examples:

Same person with multiple email addresses

Data cleaning

Process of dealing with duplicate data issues

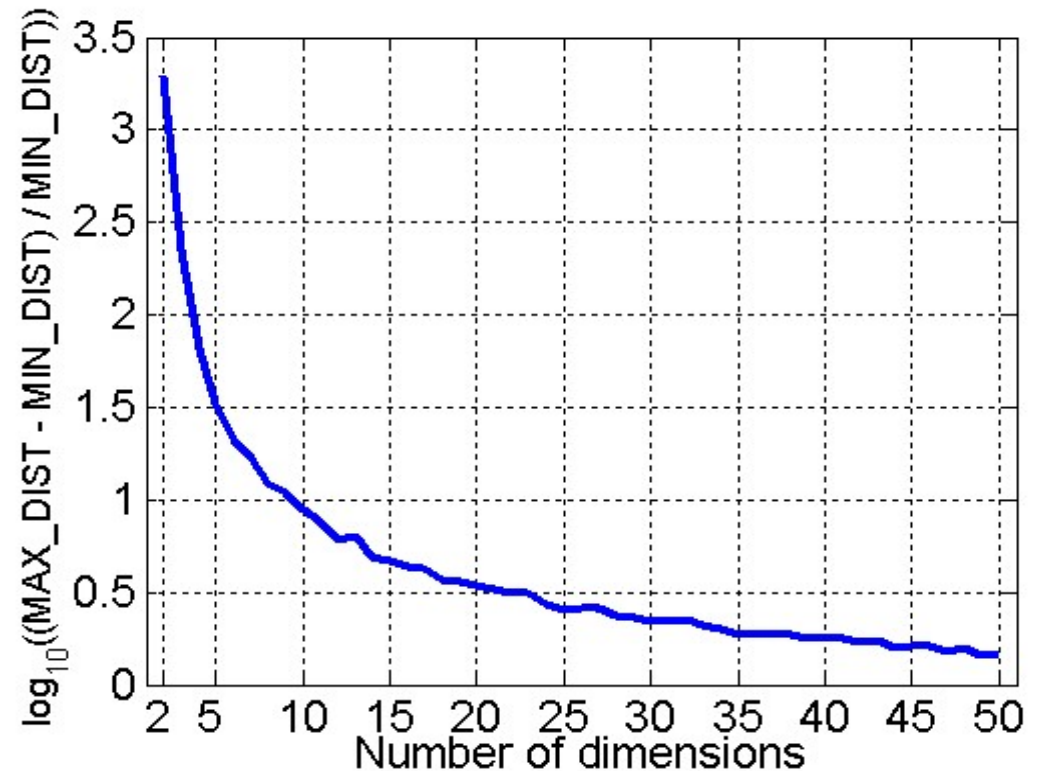
Data Preprocessing

- ❖ Aggregation
- ❖ Sampling
- ❖ Dimensionality Reduction
- ❖ Feature subset selection
- ❖ Feature creation
- ❖ Attribute Transformation
- ❖ Discretization and Binarization

Curse of Dimensionality

When dimensionality increases, data becomes increasingly sparse in the space that it occupies

Definitions of density and distance between points, which is critical for clustering and outlier detection, become less meaningful



- Randomly generate 500 points
- Compute difference between max and min distance between any pair of points

Dimensionality Reduction

Purpose:

- Avoid curse of dimensionality
- Reduce amount of time and memory required by data mining algorithms
- Allow data to be more easily visualized
- May help to eliminate irrelevant features or reduce noise

Techniques:

- Principal Component Analysis
- Singular Value Decomposition
- Others: supervised and non-linear techniques

Feature Subset Selection

Another way to reduce dimensionality of data

Redundant features

- duplicate much or all of the information contained in one or more other attributes
- Example: purchase price of a product and the amount of sales tax paid

Irrelevant features

- contain no information that is useful for the data mining task at hand
- Example: students' ID is often irrelevant to the task of predicting students' GPA

Feature Creation

Create new attributes that can capture the important information in a data set much more efficiently than the original attributes

Three general methodologies:

- Feature Extraction

 - domain-specific

- Mapping Data to New Space

- Feature Construction

 - combining features

Euclidean Distance

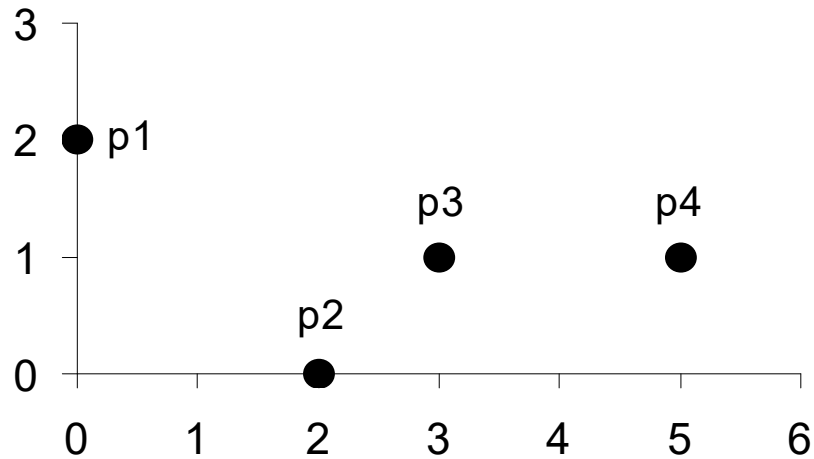
Euclidean Distance

$$\mathbf{dist} = \sqrt{\sum_{k=1}^n (\mathbf{p}_k - \mathbf{q}_k)^2}$$

Where n is the number of dimensions (attributes) and p_k and q_k are, respectively, the k^{th} attributes (components) or data objects p and q .

Standardization is necessary, if scales differ.

Euclidean Distance



point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

Distance Matrix

Minkowski Distance

Minkowski Distance is a generalization of Euclidean Distance

$$\mathbf{dist} = \left(\sum_{k=1}^n | \mathbf{p}_k - \mathbf{q}_k |^r \right)^{\frac{1}{r}}$$

Where r is a parameter, n is the number of dimensions (attributes) and p_k and q_k are, respectively, the k -th attributes (components) or data objects p and q .

Minkowski Distance: Examples

$r = 1$. City block (Manhattan, taxicab, L_1 norm) distance.

A common example of this is the Hamming distance, which is just the number of bits that are different between two binary vectors

$r = 2$. Euclidean distance

$r \rightarrow \infty$. “supremum” ($L_{\max}$ norm, L_{∞} norm) distance.

This is the maximum difference between any component of the vectors

Do not confuse r with n , i.e., all these distances are defined for all numbers of dimensions.

Minkowski Distance

point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

L1	p1	p2	p3	p4
p1	0	4	4	6
p2	4	0	2	4
p3	4	2	0	2
p4	6	4	2	0

L2	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

L_∞	p1	p2	p3	p4
p1	0	2	3	5
p2	2	0	1	3
p3	3	1	0	2
p4	5	3	2	0

Distance Matrix

Common Properties of a Distance

Distances, such as the Euclidean distance, have some well known properties.

1. $d(p, q) \geq 0$ for all p and q and $d(p, q) = 0$ only if $p = q$. (Positive definiteness)
2. $d(p, q) = d(q, p)$ for all p and q . (Symmetry)
3. $d(p, r) \leq d(p, q) + d(q, r)$ for all points p, q , and r . (Triangle Inequality)

where $d(p, q)$ is the distance (dissimilarity) between points (data objects), p and q .

A distance that satisfies these properties is a metric

Common Properties of a Similarity

Similarities, also have some well known properties.

1. $s(p, q) = 1$ (or maximum similarity) only if $p = q$.
2. $s(p, q) = s(q, p)$ for all p and q . (Symmetry)

where $s(p, q)$ is the similarity between points (data objects), p and q .

Similarity Between Binary Vectors

Common situation is that objects, p and q , have only binary attributes

Compute similarities using the following quantities

M_{01} = the number of attributes where p was 0 and q was 1

M_{10} = the number of attributes where p was 1 and q was 0

M_{00} = the number of attributes where p was 0 and q was 0

M_{11} = the number of attributes where p was 1 and q was 1

Simple Matching and Jaccard Coefficients

SMC = number of matches / number of attributes

$$= (M_{11} + M_{00}) / (M_{01} + M_{10} + M_{11} + M_{00})$$

J = number of M11 matches / number of not-both-zero attributes values

$$= (M_{11}) / (M_{01} + M_{10} + M_{11})$$

SMC versus Jaccard: Example

$$p = 1000000000$$

$$q = 0000001001$$

$M_{01} = 2$ (the number of attributes where p was 0 and q was 1)

$M_{10} = 1$ (the number of attributes where p was 1 and q was 0)

$M_{00} = 7$ (the number of attributes where p was 0 and q was 0)

$M_{11} = 0$ (the number of attributes where p was 1 and q was 1)

$$SMC = (M_{11} + M_{00}) / (M_{01} + M_{10} + M_{11} + M_{00}) = (0+7) / (2+1+0+7) = 0.7$$

$$J = (M_{11}) / (M_{01} + M_{10} + M_{11}) = 0 / (2 + 1 + 0) = 0$$

Cosine Similarity

If d_1 and d_2 are two document vectors, then

$$\cos(d_1, d_2) = (d_1 \bullet d_2) / ||d_1|| ||d_2||,$$

where $\bullet$ indicates vector dot product and $||d||$ is the length of vector d .

Example:

$$d_1 = 3 \ 2 \ 0 \ 5 \ 0 \ 0 \ 0 \ 2 \ 0 \ 0$$

$$d_2 = 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 2$$

$$d_1 \bullet d_2 = 3*1 + 2*0 + 0*0 + 5*0 + 0*0 + 0*0 + 0*0 + 2*1 + 0*0 + 0*2 = 5$$

$$||d_1|| = (3*3 + 2*2 + 0*0 + 5*5 + 0*0 + 0*0 + 0*0 + 2*2 + 0*0 + 0*0)^{0.5} = (42)^{0.5} \\ = 6.481$$

$$||d_2|| = (1*1 + 0*0 + 0*0 + 0*0 + 0*0 + 0*0 + 0*0 + 1*1 + 0*0 + 2*2)^{0.5} = (6)^{0.5} \\ = 2.245$$

$$\cos(d_1, d_2) = .3150$$

Correlation

Correlation measures the linear relationship between objects

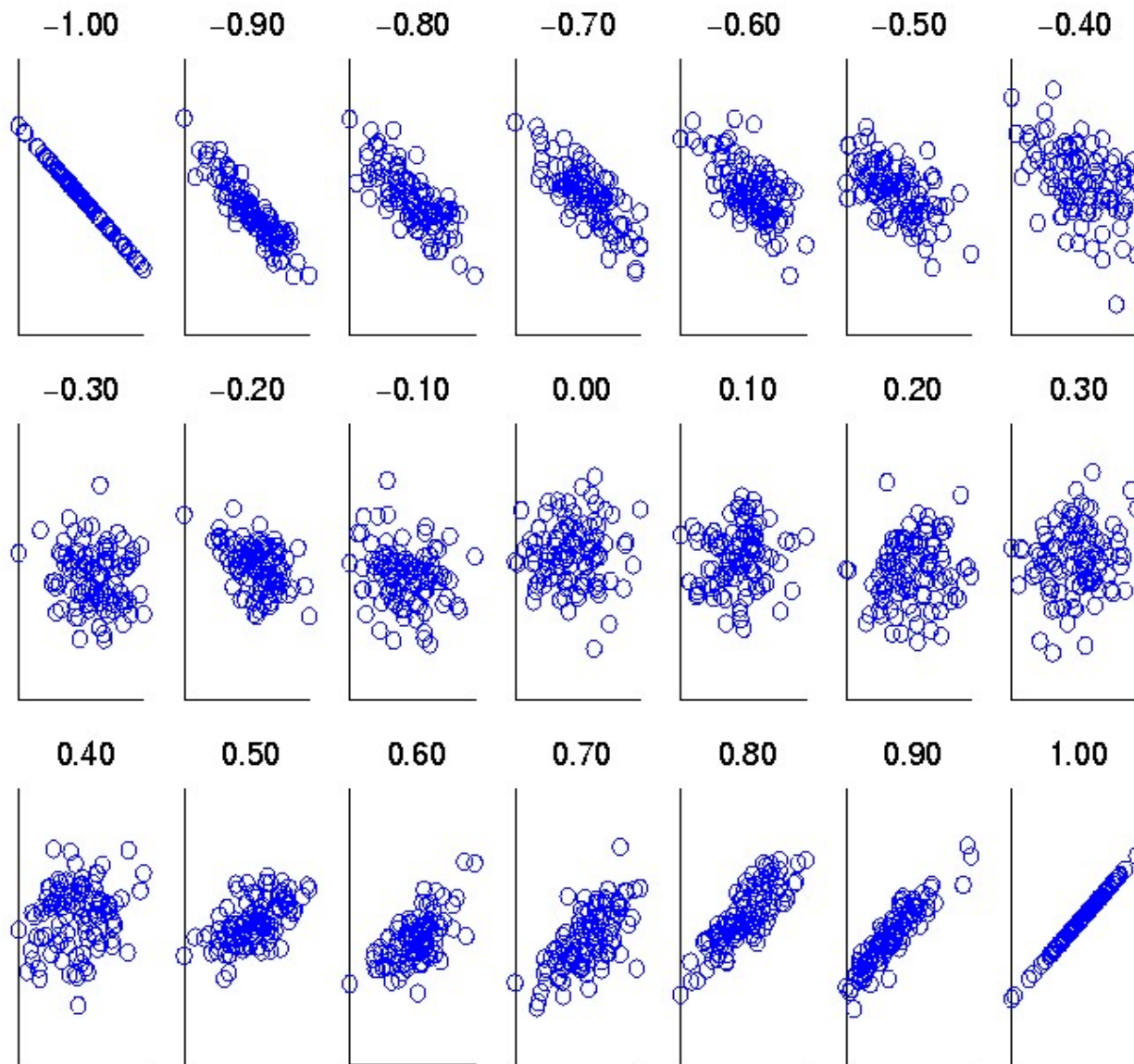
To compute correlation of vectors of size n , we standardize data objects, p and q , and then take their dot product

$$p'_k = (p_k - \text{mean}(p)) / \text{std}(p)$$

$$q'_k = (q_k - \text{mean}(q)) / \text{std}(q)$$

$$\text{correlation}(p, q) = \frac{1}{n} (p' \bullet q')$$

Visually Evaluating Correlation



Scatter plots showing the similarity from -1 to 1.

Exercise
